

EXERCISE

Solve the following system of differential equations:

1. $\frac{dx}{dt} = -\omega y, \frac{dy}{dt} = \omega x,$
2. $\frac{dx}{dt} + 2x - y = 0, \frac{dy}{dt} - x + 2y = 0,$
3. $\frac{dx}{dt} - 3x - 2y = 0, \frac{dy}{dt} + 5x + 3y = 0,$
4. $\frac{dx}{dt} + 2x + 3y = 0, \frac{dy}{dt} + 3x + 2y = 2e^{2t},$
5. $\frac{dx}{dt} + 5x + y = e^t; \frac{dy}{dt} - x + 3y = e^{2t},$
6. $\frac{dx}{dt} + x + 2y = e^t; \frac{dy}{dt} + 2x + y = 3e^t,$
7. $\frac{dx}{dt} + 2\frac{dy}{dt} - 2x + 2y = 3e^t; 3\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 4e^{2t},$
8. $\frac{dx}{dt} = ax + by, \frac{dy}{dt} = bx + ay,$
9. $\frac{dx}{dt} = ax + by, \frac{dy}{dt} = cx + dy,$
10. $\frac{dx}{dt} - 7x + y = e^t, \frac{dy}{dt} - 2x - 5y = 2,$
11. $\frac{dx}{dt} + 2x - 3y = t; \frac{dy}{dt} - 3x + 2y = e^{2t},$
12. $\frac{dy}{dx} + y = z + e^x; \frac{dz}{dx} + z = y + e^x,$
13. $\frac{dx}{dt} + \frac{dy}{dt} + 2y = 0, \frac{dx}{dt} - 3x - 2y = 0,$
14. $\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 0, \frac{dy}{dt} + 5x + 3y = 0,$
15. $4\frac{dx}{dt} + 9\frac{dy}{dt} + 2x + 31y = e^t; 3\frac{dx}{dt} + 7\frac{dy}{dt} + x + 24y = 3,$
16. $\frac{dx}{dt} - \frac{dy}{dt} - y = -e^t; \frac{dy}{dt} + x - y = e^{2t},$
17. $\frac{dx}{dt} + \frac{dy}{dt} - x + 3y = e^{-t} - 1; \frac{dx}{dt} + \frac{dy}{dt} + 2x + y = e^{2t} + t,$
18. $\frac{dx}{dt} + \frac{dy}{dt} - x = 2t + 1; 2\frac{dx}{dt} + 2\frac{dy}{dt} + x = t,$
19. $3\frac{dy}{dt} - \frac{dx}{dt} + y = e^t, 3\frac{dx}{dt} - \frac{dy}{dt} + x = 0,$
20. $\frac{d^2x}{dt^2} - 3x - 4y = 0, \frac{d^2y}{dt^2} + x + y = 0,$
21. $\frac{d^2x}{dt^2} - 2\frac{dy}{dt} - x = e^t \cos t, \frac{d^2y}{dt^2} + 2\frac{dx}{dt} - y = e^t \sin t,$
22. $\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} + \frac{dx}{dt} + x + y = e^t, \frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} + \frac{dx}{dt} = e^{-t},$
23. $\frac{d^2x}{dt^2} + a^2y = 0, \frac{d^2y}{dt^2} - a^2x = 0,$

24. $\frac{dx}{dt} + \frac{dy}{dt} + x - y = e^t$; $\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} + \frac{dx}{dt} - \frac{dy}{dt} + x + y = t^2$,
25. $\frac{dx}{dt} = x^2 + xy$, $\frac{dy}{dt} = y^2 + xy$,
26. $\frac{dx}{dt} + 2x - 7y = 0$, $\frac{dy}{dt} - 3x - 2y = 0$, given that $x(0) = 9$, $y(0) = -1$,
27. $\frac{dx}{dt} + 2x + 8y = 0$, $\frac{dy}{dt} - x - 6y = 0$, given that $x(0) = 4$, $y(0) = 1$,
28. $\frac{dx}{dt} - 6x + 4y = 0$, $\frac{dy}{dt} - x - 2y = 0$, given that $x(0) = 2$, $y(0) = 3$,
29. $\frac{d^2x}{dt^2} + \frac{dy}{dt} + x + y = t$, $\frac{dy}{dt} + 2x + y = 0$, given that $x(0) = y(0) = 0$,
30. $\frac{d^2y}{dt^2} - 16x = t$, $\frac{d^2x}{dt^2} - y = 1$, given that $x = y = 0$, $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = \frac{1}{4}$ at $t = 0$,
31. $t \frac{dx}{dt} + y = 0$, $t \frac{dy}{dt} + x = 0$,
32. $t \frac{dx}{dt} + 2x - 2y = t$, $t \frac{dy}{dt} + x + 5y = t$,
33. Show that solutions of the equations $\frac{dx}{dt} = -2y$, $\frac{dy}{dt} = x$ is given by $x^2 + 2(y^2 + 0)$.

ANSWERS

1. $x = A \cos \omega t + B \sin \omega t$; $y = A \sin \omega t - B \cos \omega t$,
2. $x = Ae^{-t} + Be^{-3t}$, $y = Ae^{-t} - Be^{-3t}$,
3. $x = A \cos t + B \sin t$; $y = \frac{1}{2}(B - 3A) \cos t - \frac{1}{2}(A + 3B) \sin t$,
4. $x = Ae^t + Be^{-5t} - \frac{6}{7}e^{2t}$; $y = -Ae^t + Be^{-5t} + \frac{8}{7}e^{2t}$,
5. $x = (A + Bt)e^{-4t} + \frac{4}{25}e^t - \frac{1}{36}e^{2t}$; $y = -[A + (1 + t)B]e^{-4t} + \frac{1}{25}e^t + \frac{7}{36}e^{2t}$,
6. $x = Ae^t + Be^{-3t} - te^t$, $y = -Ae^t + Be^{-3t} + e^t + te^t$,
7. $x = \frac{1}{2}e^{2t} - \frac{3}{11}e^t + Ae^{-\frac{6t}{5}}$; $y = \frac{15}{22}e^t + Be^{-t} - 8Ae^{-\frac{6t}{5}}$,
8. $x = Ae^{(a+b)t} + Be^{(a-b)t}$; $y = Ae^{(a+b)t} - Be^{(a-b)t}$,
9. $x = Ae^{m_1 t} + Be^{m_2 t}$; $y = \frac{1}{b}[(m_1 - a)Ae^{m_1 t} + (m_2 - a)Be^{m_2 t}]$, where
- $$m_{2,1} = \frac{1}{2} \left\{ a + d \pm \sqrt{(a-d)^2 + 4bc} \right\},$$

$$10. x = (A \cos t + B \sin t) e^{6t} - \frac{2}{13} e^t - \frac{2}{37},$$

$$y = [(A - B) \cos t + (A + B) \sin t] e^{6t} + \frac{1}{13} e^t - \frac{14}{37},$$

$$11. x = Ae^{-5t} + Be^t + \frac{3}{7} e^{2t} - \frac{2t}{5} - \frac{13}{25}; y = -Ae^{-5t} + Be^t + \frac{4}{7} e^{2t} - \frac{3t}{5} - \frac{12}{25},$$

$$12. y = A + Be^{-2x} + e^x; z = A - Be^{-2x} + e^x,$$

$$13. x = Ae^{-3t} + Be^{2t}, y = -3Ae^{-3t} - \frac{1}{2} Be^{2t},$$

$$14. x = A \cos t + B \sin t; y = \frac{1}{2}(B - 3A) \cos t - \frac{1}{2}(A + 3B) \sin t,$$

$$15. x = (A \sin t + B \cos t) e^{-4t} + \frac{31}{26} e^t - \frac{93}{17};$$

$$y = [(A - B) \sin t - (A + B) \cos t] e^{-4t} - \frac{2}{13} e^t + \frac{6}{17},$$

$$16. x = (A - B) \cos t + (A + B) \sin t + \frac{3}{5} e^{2t}; y = A \cos t + B \sin t + \frac{2}{5} e^{2t} + \frac{1}{2} e^t,$$

$$17. x = 2Ae^{\frac{7t}{5}} + \frac{5}{17} e^{2t} + \frac{3t}{7} - \frac{1}{49}; y = 3Ae^{\frac{7t}{5}} - \frac{1}{17} e^{2t} + \frac{1}{2} e^{-t} + \frac{t}{7} - \frac{26}{49},$$

$$18. x = -t - \frac{2}{3}, y = \frac{t^2}{2} + \frac{4}{3} t + A,$$

$$19. x = Ae^{-\frac{1}{2}t} + Be^{-\frac{1}{4}t} + \frac{1}{15} e^t, y = Ae^{-\frac{1}{2}t} - Be^{-\frac{1}{4}t} + \frac{4}{15} e^t,$$

$$20. x = (A + Bt) e^t + (C + Dt) e^{-t}; y = \frac{1}{2}[B(1 - t) - A] e^t - \frac{1}{2}[D(1 + t) + C] e^{-t},$$

$$21. x = (c_1 + c_2 t) \cos t + (c_3 + c_4 t) \sin t + \frac{e^t}{25} (4 \sin t - 3 \cos t);$$

$$y = -(c_1 + c_2 t) \sin t + (c_3 + c_4 t) \cos t + \frac{e^t}{25} (3 \sin t + 4 \cos t),$$

$$22. x = -e^t - 2e^{-t} - A; y = 2e^t + e^{-t} + A,$$

$$23. x = e^{\frac{at}{\sqrt{2}}} \left(A \cos \frac{at}{\sqrt{2}} + B \sin \frac{at}{\sqrt{2}} \right) + e^{-\frac{at}{\sqrt{2}}} \left(C \cos \frac{at}{\sqrt{2}} + D \sin \frac{at}{\sqrt{2}} \right)$$

$$y = e^{\frac{at}{\sqrt{2}}} \left(A \sin \frac{at}{\sqrt{2}} - B \cos \frac{at}{\sqrt{2}} \right) + e^{-\frac{at}{\sqrt{2}}} \left(D \cos \frac{at}{\sqrt{2}} - C \sin \frac{at}{\sqrt{2}} \right).$$

$$24. x = \frac{1}{2}(e^t - 2t + t^2); y = \frac{1}{2}(2t + t^2 - 3e^t),$$

$$25. \quad x = -\frac{1}{(1+c_1)t+c_2}; \quad y = -\frac{c_1}{(1+c_1)t+c_2},$$

$$26. \quad x = 2e^{5t} + 7e^{-5t}, \quad y = 2e^{5t} - 3e^{-5t},$$

$$27. \quad x = 4e^{4t}[\cos 2t - 2\sin 2t], \quad y = e^{4t}[\cos 2t + 3\sin 2t],$$

$$28. \quad x = 2e^{4t} - 8te^{4t}, \quad y = 3e^{4t} - 4te^{4t},$$

$$29. \quad x = -2e^t + 2e^{-t} - t, \quad y = 2(e^t - 2et^{-1} + t - 1),$$

$$30. \quad x = \frac{1}{16}(3e^{2t} - e^{-2t}) + \frac{9}{32}\sin 2t - \frac{1}{8}\cos 2t - \frac{t}{16},$$

$$y = \frac{1}{4}(3e^{2t} - e^{-2t}) - \frac{9}{8}\sin 2t + \frac{1}{2}\cos 2t - 1,$$

$$31. \quad x = c_1t + \frac{c_2}{t}; \quad y = -c_1t + \frac{c_2}{t},$$

$$32. \quad x = \frac{A}{t^3} + \frac{B}{t^4} + \frac{3}{10}t + \frac{1}{15}t^2; \quad y = -\frac{A}{2t^3} - \frac{B}{t^4} - \frac{1}{20}t + \frac{2}{15}t^2.$$